

Understanding DevOps: A Revolution in Software Development

In the field of IT and software development, DevOps implementation is gaining significant popularity. This article takes a quick look at the important concepts of DevOps and the phases of its life cycle. It also maps the relevant open source tools, and finally, highlights how it can bring value to the IT industry.

The word DevOps is a combination of development and operations. DevOps is not software — not a tool, not a product and not a programming language, but an approach whereby the development and operations teams work together, instead of waiting for the other to finish tasks, to enable continuous delivery of value to end users. This software development methodology improves the collaboration between the development and the operations teams using various automation tools. These tools are implemented during different phases, which are a part of the DevOps life cycle.

DevOps is gaining popularity

DevOps is gaining popularity because it bridges the gap between developers and the supporting operations team. The DevOps approach aims at:

- Delivering high-quality software in a shorter development life cycle.
- Deployment in frequent cycles.
- Reducing the time to move into production, from conceptualising an idea.
- Accelerating application delivery across enterprise portfolios.
- Enabling rapid building and delivery of products.
- Delivering applications and services at high velocity.

Various stages in the DevOps life cycle

An understanding of DevOps is incomplete without having knowledge of its life cycle. The various phases that comprise the DevOps life cycle are highlighted in Figure 1 and described below.

- 1. Continuous development:** The first phase of the DevOps life cycle involves ‘planning’ and ‘coding’ of the software. Planning includes activities like designing the blueprint of the module, and identifying the resources and the algorithm to be used. Once the plan is finalised, the developers code the application and maintain it using popular tools like Git, Gradle and Maven.
- 2. Continuous testing:** To catch any error and ensure the reliability of the software, the developed modules are continuously tested for bugs. In this phase, testing tools like Selenium (Se), TestNG, JUnit, etc, are used.
- 3. Continuous integration:** This phase is the heart of the DevOps life cycle in which code supporting new functionality in the Git repository is continuously

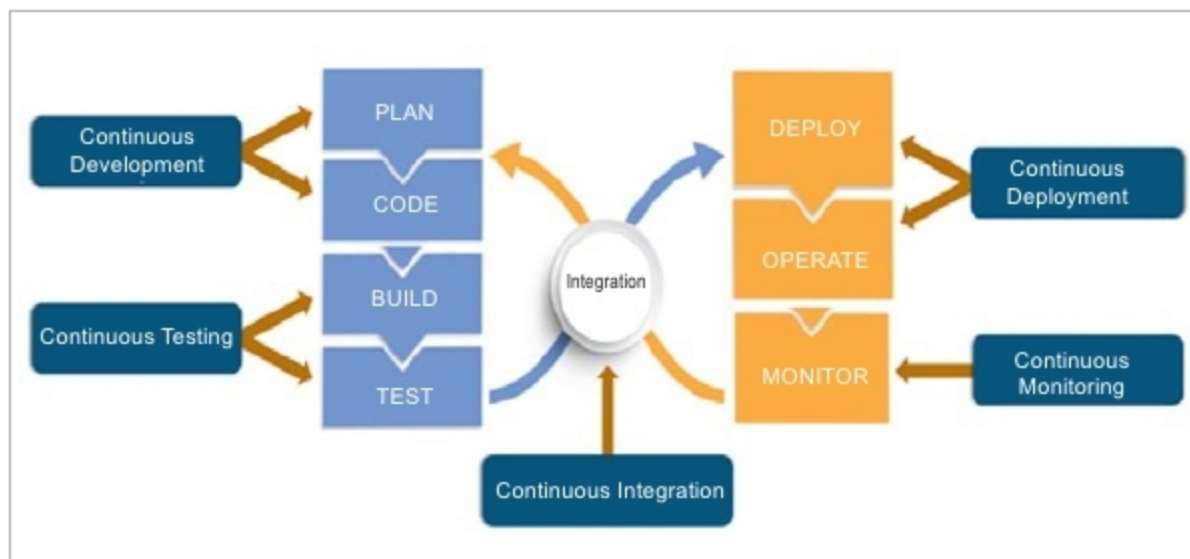


Figure 1: Different phases of the DevOps life cycle

